Yonglan Liu

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Education

The University of Akrons, Chemical Engineering — Akron, OH, USA
Jan 2017 – May 2021

Chongqing University, Bioengineering — Chongqing, China
Sept 2010 – June 2014

Experience

Guidehouse, Senior Technical Analyst | Computational Chemist — Bethesda, MD, US
Mar 2025 – present
1 year

Senior Technical Analyst at Guidehouse and serving as a Computational Chemist at NIH, leverage different computational technicals for drug discovery.

- Led the design and operationalization of integrated, end-to-end discovery workflows combining docking, MD-based refinement, alchemical free energy perturbation (FEP), and ML-driven developability prediction to guide compound prioritization across NIH-funded programs.
- Owned the architecture and reproducibility standards for alchemical FEP pipelines using OpenMM and MBAR, enabling quantitative protein–ligand binding affinity prediction and decision support for structure-based lead optimization.
- Curated and analyzed target-focused chemical libraries to support iterative design–test–learn cycles, partnering with medicinal chemists to leverage free-energy–informed SAR analysis for hypothesis refinement and compound selection.
- Applied structure-based, ligand-based, and fragment-based design strategies to manage trade-offs among potency, selectivity, and developability constraints.
- Served as a technical bridge between computation and experimental teams, translating FEP trends, MD-derived conformational insights, virtual screening, and machine learning results into actionable hypotheses that directly informed synthesis and experimental prioritization.

National Cancer Institute, National Institute of Health, Postdoctoral Fellow | Research Fellow — Frederic, MD, US
May 2021 – Feb 2025
3 years 10 months

- Independently led multiple structure- and mechanism-driven discovery efforts in cancer signaling, driving hypothesis formation, computational strategy, and interpretation across protein–ligand and protein–protein interaction systems.
- Developed reusable computational pipelines integrating MD-based conformational sampling with free-energy analyses (MM/GBSA, FEP) to enable systematic evaluation across multiple targets and ligand series.
- Identified cryptic and allosteric binding pockets via MD-driven conformational exploration, informing inhibitor and PROTAC design strategies.
- Applied AI-based protein structure prediction tools (AlphaFold, RoseTTAFold) to generate working models for proteins and protein–protein complexes in the absence of experimental structures, enabling downstream modeling and interaction analysis.

The University of Akron, Research Assistant | Akron, OH, US | Jan 2017 – May 2021 | 4 years 5 months

- Built computational chemistry and biology pipelines to model protein–protein, protein–ligand, protein–membrane, and peptide interactions, supporting small-molecule and peptide-based therapeutic design.
- Applied molecular docking and MD simulations to characterize binding mechanisms and conformational behavior, guiding structure- and sequence-level optimization.
- Developed ML/AI models to predict sequence/structure–property relationships for peptides, small molecules, and biofunctional materials, working closely with experimental collaborators for validation.

Skills

Programming & Scientific Computing

Molecular Modeling & Simulation & Cheminformatics

Machine Learning & Deep Learning

Languages

Chinese (Mandarin)
Native speaker

English
Fluent

Interests

Data Science & Artificial Intelligence

Drug Discovery & Computational Chemistry

Drug Design & Structural Biology

Certificates

Data Scientist

Natural Language Processing (NLP) with Attention Models

Natural Language Processing (NLP) with Probabilistic Models

Natural Language Processing (NLP) with Sequence Models

Natural Language Processing (NLP) with Classification and Vector Spaces

Convolutional Neural Networks (CNN)

Projects

End-to-End Alchemical Free Energy Perturbation (FEP) Pipeline for Ligand Binding Affinity Prediction

- Designed and implemented an end-to-end alchemical FEP pipeline using OpenMM, openmmtools, and MBAR, enabling automated prediction of ligand binding affinities in real protein–ligand discovery projects.
- Built a reproducible workflow covering protein preparation, ligand mapping, alchemical transformations, solvation, equilibration, production MD, and free energy analysis, supporting consistent and auditable computational decision-making.
- Developed Python utilities and a lightweight dashboard to visualize $\Delta\Delta G$ values, uncertainty estimates, and simulation QC metrics, enabling rapid compound comparison and prioritization across ligand series in real design cycles.
- Applied the pipeline to real medicinal chemistry use cases, demonstrating its value in guiding structure-based lead optimization and reducing reliance on heuristic scoring functions.

Machine Learning–Driven Peptide Design & Experimental Validation

- Built ML models to identify and design self-assembling peptides with anti-amyloid properties, targeting amyloid-related diseases including Alzheimer's disease and type 2 diabetes.
- Represented peptide physicochemical properties using NNAAIndex-based descriptors and performed feature analysis to identify determinants of aggregation behavior.
- Trained and optimized SVM models (Python, GridSearchCV) to discriminate self-assembling versus non-assembling peptides.
- Collaborated with experimental teams to validate predictions, resulting in five confirmed peptides with strong anti-amyloid activity, including one patented peptide [Authorized patent (ZL201410100022.5)].

Protein Kinase Structural Selectivity Analytics

- Curated and analyzed 6,700+ protein–ligand complex structures from the PDB to characterize structural determinants of binding site accessibility.
- Developed structural clustering and network-based analytics to differentiate drug-accessible regions across related protein families.
- Contributed to development of the KDS (Kinase Drug Selectivity) software platform.

References

Professor Ruth Nussinov

Professor Jie Zheng